

Lecture 3: Glass Optics, Transfer Matrices, Transverse Focusing

Glass optics examples

- ▶ Good introduction to transfer map methods
- ▶ Intuitive: lots of good everyday examples
- ▶ Rotational symmetry: easy treatment, two variables
 - ▶ x – position of a ray
 - ▶ m – its slope
- ▶ Transfer map: describes the effect of the system on the initial coordinates x and m :

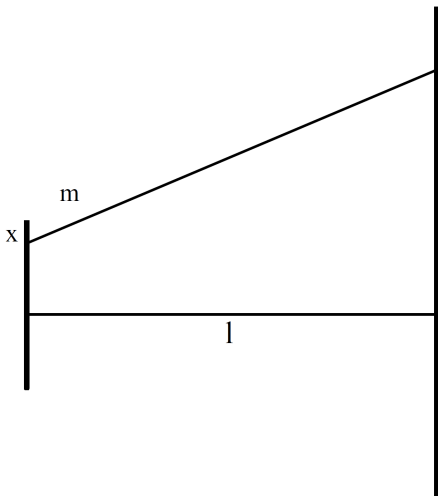
$$\begin{pmatrix} x_2 \\ m_2 \end{pmatrix} = \mathcal{M} \begin{pmatrix} x_1 \\ m_1 \end{pmatrix}$$

- ▶ If we restrict ourselves to linear motion: transfer map $\Rightarrow$ *transfer matrix*:

$$M = \begin{pmatrix} (x|x) & (x|m) \\ (m|x) & (m|m) \end{pmatrix}$$



$$\begin{pmatrix} x_{n+1} \\ m_{n+1} \end{pmatrix} = M_n \left(\cdots \left(M_1 \begin{pmatrix} x_1 \\ m_1 \end{pmatrix} \right) \cdots \right) = (M_n \cdot \dots \cdot M_1) \begin{pmatrix} x_1 \\ m_1 \end{pmatrix}$$



- ▶ Equations of motion:

$$x_2 = x_1 + m_1 \cdot l$$

$$m_2 = m_1$$

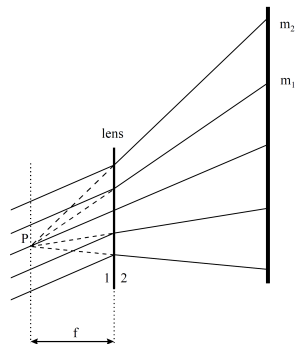
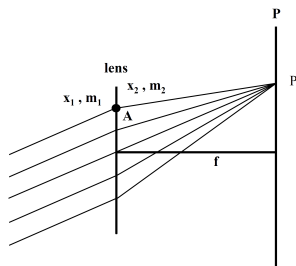


$$\begin{pmatrix} x_2 \\ m_2 \end{pmatrix} = \begin{pmatrix} 1 & l \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ m_1 \end{pmatrix}$$

- ▶ Transfer matrix depends only on the properties of the element (“lattice parameters”)
- ▶ Vector $\begin{pmatrix} x_1 \\ m_1 \end{pmatrix}$ depends only on the parameters of the ray (“beam”)
- ▶ $\det(M) = 1$
- ▶ What would be the matrix for a combination of drifts l_1 and l_2 ?

Thin lens

- ▶ Changes direction of the light ray
- ▶ Thin lens (“kick”) approximation: no length
- ▶ Positions are not changed, but directions are
- ▶ A ray hitting the center of the lens goes straight through
- ▶ For the focusing lens: any bundle of *parallel* light is unified at a single point a distance f after the lens
- ▶ For the defocusing lens: any bundle of parallel light exits the lens in such a way that it appears to come from a point a distance f in front of the lens
- ▶ f is called the focal length



Focusing lens

- ▶ Equations of motion:

$$x_2 = x_1$$

$$p = f \cdot m_1$$

$$x_1 + m_2 \cdot f = p$$



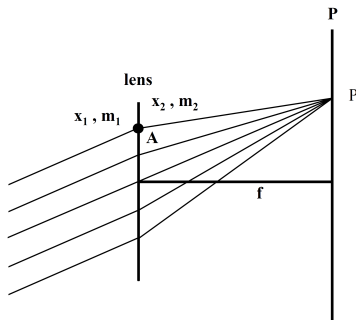
$$x_2 = x_1$$

$$m_2 = -\frac{x_1}{f} + m_1$$



$$\begin{pmatrix} x_2 \\ m_2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -\frac{1}{f} & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ m_1 \end{pmatrix}$$

- ▶ Again, the matrix depends only on the lens, the coordinate vector – only on the ray
- ▶ $\det M = 1$
- ▶ What is the matrix of the combination of two lenses with focusing distances f_1 and f_2 ?



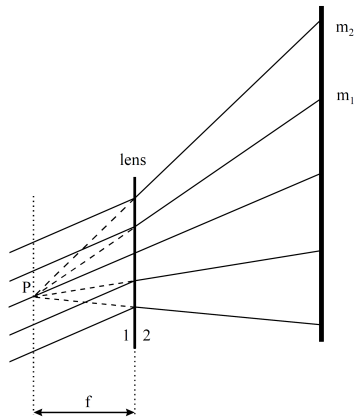
Defocusing lens

► Equations of motion:

$$\begin{aligned}x_2 &= x_1 \\p &= -f \cdot m_1 \\x_1 - m_2 \cdot f &= p\end{aligned}$$

$$\begin{aligned}x_2 &= x_1 \\m_2 &= +\frac{x_1}{f} + m_1\end{aligned}$$

$$\begin{pmatrix} x_2 \\ m_2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ +\frac{1}{f} & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ m_1 \end{pmatrix}$$

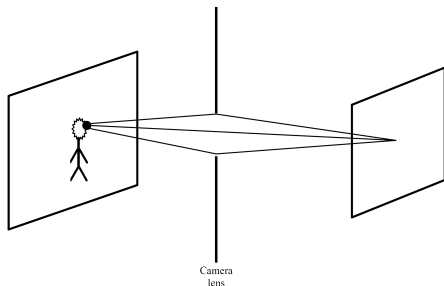


Imaging (point-to-point) system

- ▶ What are the examples of imaging systems you can think of?
- ▶ Key properties to recreate the image of an object:
 - ▶ All the light reflecting from a certain point on an object has to be reunited at one spot on the screen
 - ▶ Hence, the final position of a ray is independent of its initial angle, only depends on the initial position

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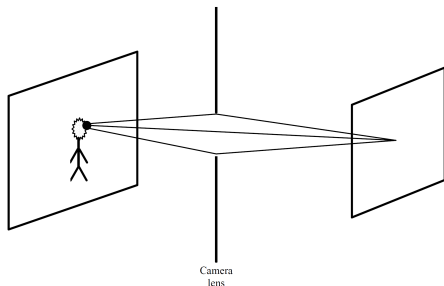


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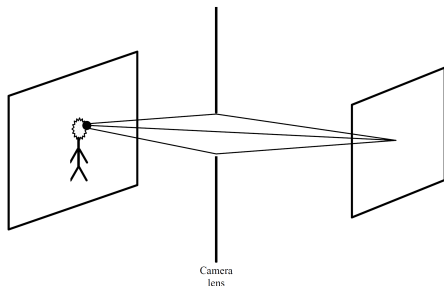


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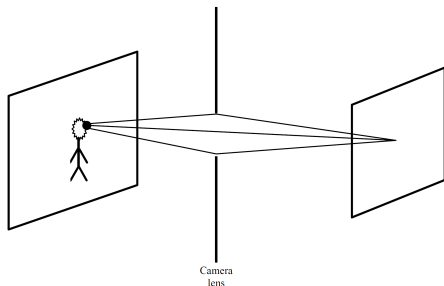


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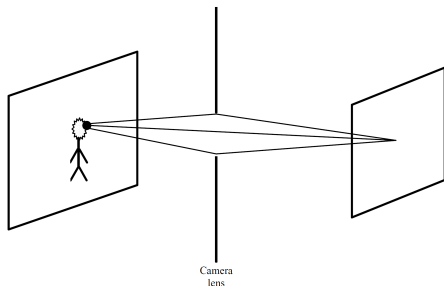


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- ▶ Is x-ray machine an example of



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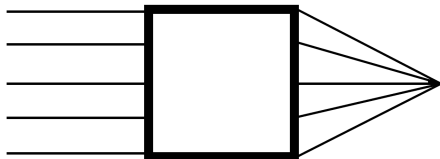
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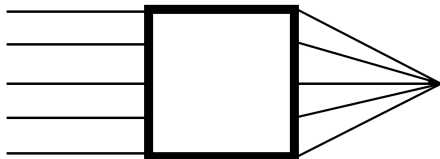
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Parallel-to-point



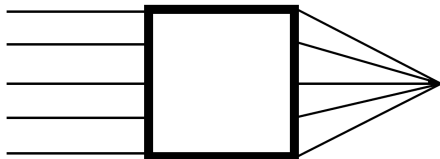
► Examples:

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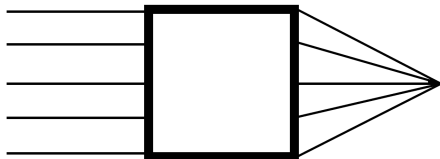
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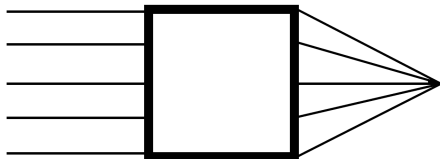
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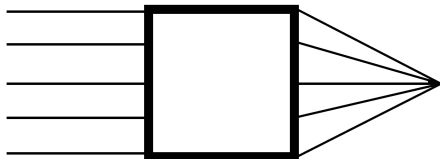
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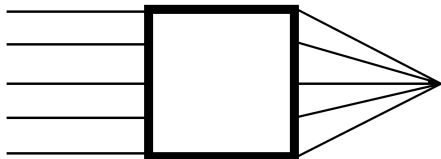
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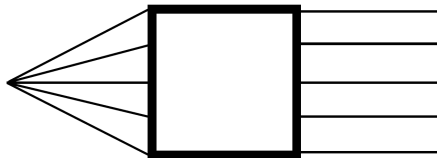
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- ▶ $(x|x) = 0$

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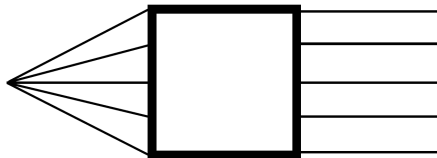
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- ▶ DLD: $1 - l_2/f = 0, \Rightarrow l_2 = f, l_1$ arbitrary

Point-to-parallel



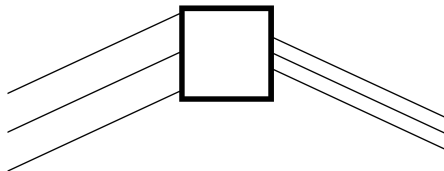
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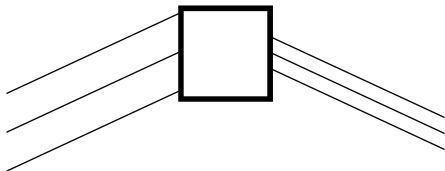
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Parallel-to-parallel system



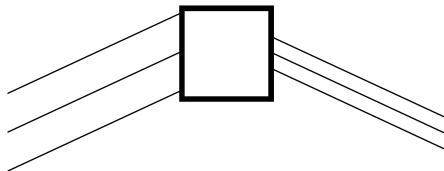
► Example:

Parallel-to-parallel system



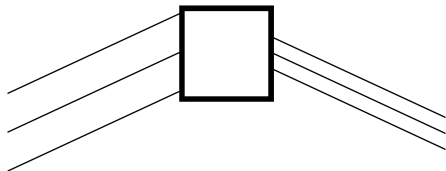
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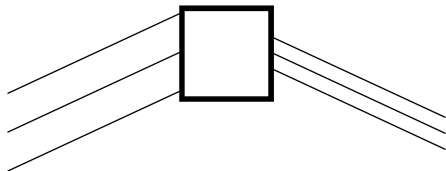
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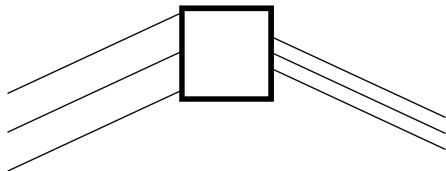
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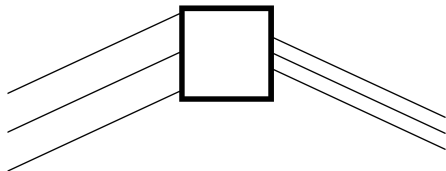
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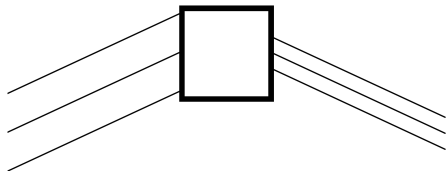
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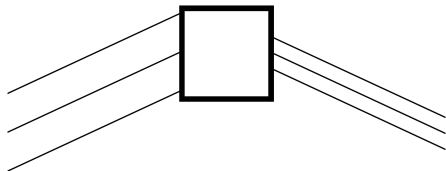
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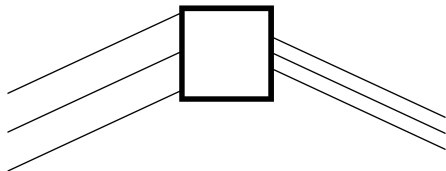


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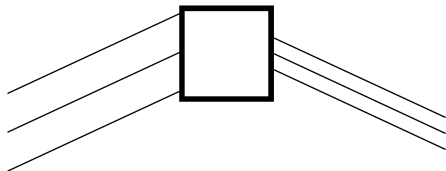


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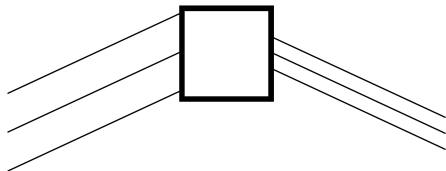


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- ▶ $(m|m) = -\frac{f_1}{f_2}$

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- ▶ In many cases, designing a segment of an accelerator lattice involves the same technique

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 - ▶ ...so it is good to have a basic level intuition on how to work with them
- ▶ We will cover this topic next, and looking at familiar/simple glass optics appears to be an easy segue into it
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 - ▶ ...so we have to squeeze the beam in one direction then the opposite
- ▶ In the global sense: accelerators are huge imaging systems.

Steering and Focusing Charged Particles

Steering and Focusing Charged Particles I

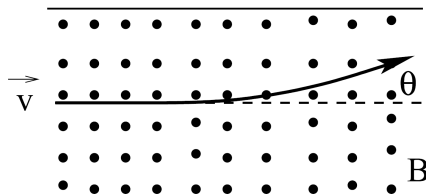
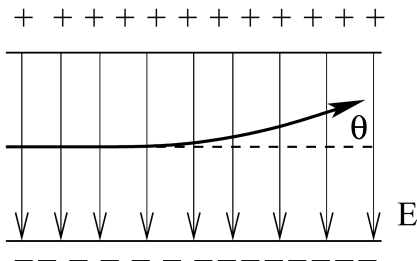
- ▶ In order to form the beam of particles and keep it tight for the duration of the acceleration cycle, we need three basic types of elements:
 - ▶ Guiding elements: keep the particles on a well-defined *reference orbit*
 - ▶ Focusing elements: keep the particles within the vacuum chamber close to the reference orbit
 - ▶ Accelerating elements: give the particles a boost to get to the design energy

Steering and Focusing Charged Particles II

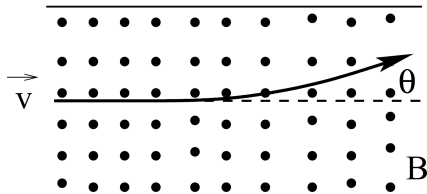
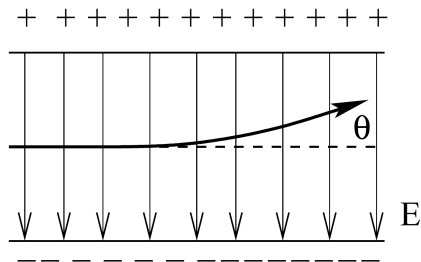
- Assume we have a charged particle, and we need to choose whether we use E or B to steer off its straight track:

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$$

- What would be more efficient/appropriate?



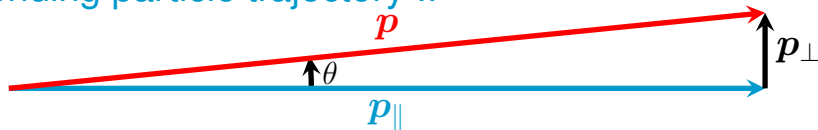
Bending particle trajectory I



Algorithm:

- ▶ Consider paraxial approximation (appropriate in most cases)
- ▶ Derive expression for small angle change as the particle steps through the E or B field
- ▶ Integrate to find the overall change in angle per unit length
- ▶ Substitute E or B into the resulting expression
- ▶ Compare the outcomes and make conclusions about the suitability

Bending particle trajectory II



- ▶ $p_{\parallel} \gg p_{\perp}$ ($p_{\perp}$ is exaggerated in the picture), so θ is small:

$$\tan(\theta) \approx \sin(\theta) \approx \theta \approx \frac{p_{\perp}}{p_{\parallel}} \approx \frac{p_{\perp}}{p}$$

- ▶ Denote small change in transverse momentum dp , then

$$d\theta = \frac{dp}{p} = \frac{1}{p} \frac{dp}{dt} \frac{dt}{dz} dz = \frac{1}{p} F \frac{1}{v} dz$$

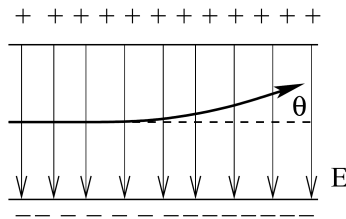
- ▶ Thus, the total angle change

$$\theta = \frac{1}{pv} \int_0^L F dz$$

Bending particle trajectory III

$$\theta = \frac{1}{pv} \int_0^L F dz$$

Let's substitute some numbers: assume we deal with electrons with total energy 10 GeV, and we want to bend it through 0.01 rad over 1 m. In this case relativistic $\beta = 0.995$ and energy over charge is $10 \text{ GeV}/e = 10^{10} \text{ V}$.



- ▶ Let's assume the particle overall direction of motion is $\hat{z}$, and we want to bend the particle in the $\hat{x}$ direction. Then $\mathbf{E}$ should be in the $-\hat{x}$ direction, and $\mathbf{B}$ – in the $\hat{y}$ direction (verify that!)

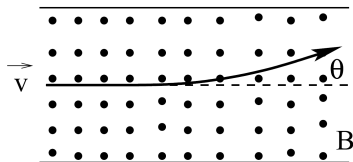
- ▶ If we use $\mathbf{E}$: $F_E \hat{x} = -eE_x(-\hat{x}) = eE_x \hat{x}$ and $\theta_{\text{total}} = \frac{1}{pv} \int_0^L eE_x dz$

$$E_x L = \int_0^L E_x dz = \frac{\theta pv}{e} = \frac{\theta(\gamma mc^2)\beta^2}{e} = (10 \text{ mrad})(10^{10} \text{ V})(0.99) = 99 \text{ MV}$$

Bending particle trajectory IV

$$\theta = \frac{1}{pv} \int_0^L F dz$$

Let's substitute some numbers: assume we deal with electrons with total energy 10 GeV, and we want to bend it through 0.01 rad over 1 m. In this case relativistic $\beta = 0.995$ and energy over charge is $10 \text{ GeV}/e = 10^{10} \text{ V}$.



- ▶ Let's assume the particle overall direction of motion is $\hat{z}$, and we want to bend the particle in the $\hat{x}$ direction. Then $\mathbf{E}$ should be in the $-\hat{x}$ direction, and $\mathbf{B}$ – in the $\hat{y}$ direction (verify that!)

- ▶ Same for $\mathbf{B}$: $F_B \hat{x} = -ev\hat{z} \times B_y \hat{y} = evB_y \hat{x}$ and $\theta_{\text{total}} = \frac{1}{pv} \int_0^L evB_y dz$

$$B_y L = \int_0^L B_y dz = \frac{\theta p}{e} = \frac{\theta(\gamma mc^2)\beta}{ec} = \frac{(10 \text{ mrad})(10^{10} \text{ V})(0.995)}{3 \times 10^8 \text{ m/s}} = 0.33 \text{ T}$$

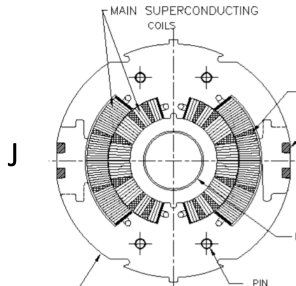
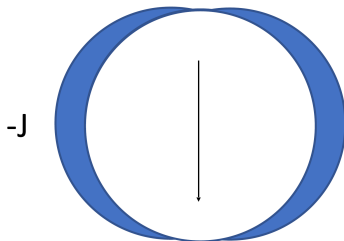
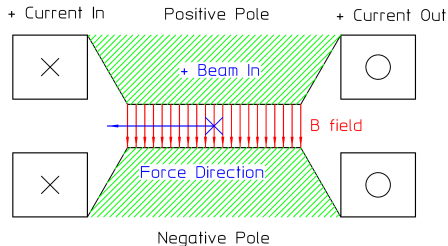
- ▶ Summary: $E_x = 99 \text{ MV/m}$ is tough while $B_y = 0.33 \text{ T/m}$ is reasonable.

Magnets

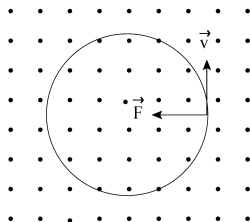
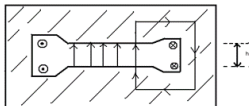
- ▶ So we agreed: we bend and focus using the magnetic field
- ▶ But there is a catch: magnetic forces do no work ($\mathbf{F} \perp \mathbf{v}$) $\Rightarrow$ use electric field for acceleration
- ▶ Different magnet shapes:
 - ▶ Dipole
 - ▶ Quadrupole
 - ▶ Sextupole
 - ▶ N-pole
 - ▶ Combined function
 - ▶ Solenoid
 - ▶ ...
- ▶ Different materials:
 - ▶ Iron-dominated: shaped by the iron
 - ▶ Superconducting: shaped by coil placement
 - ▶ Superferric: combination of both

Bending particles: dipole magnet

- ▶ Two poles, constant field, steers particle beam



Dipole magnet field and motion

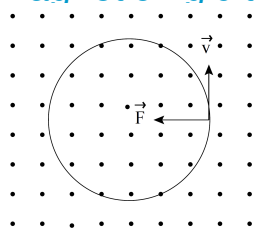


- ▶ The field in the gap near the axis (where the beam is) is nearly uniform.
 - ▶ Field calculation = try it at home (hint: in the next few slides I'll show how to do this for a quadrupole magnet). Your answer should be

$$B_0 = \frac{\mu_0 n I}{h}, \text{ where } n - \# \text{ of turns in the coil, } I - \text{coil current, } h - \text{size of the gap}$$

- ▶ Uniform magnetic field $\Rightarrow$ circular motion
 - ▶ ...and it is intrinsically unstable
 - ▶ That's where combined-function magnet helps, but let's look at the quad first

Magnetic rigidity



- ▶ Let's derive an important relation from the equation of motion in the uniform magnetic field
- ▶ Two forces are in balance: the Lorentz force $F_L = evB$ ($v \perp B$) and the centrifugal force $F = \gamma m_0 v^2 / \rho$, where ρ is the radius of curvature
- ▶ Thus,

$$F = \gamma m_0 \frac{v^2}{\rho} = qvB$$



$$\boxed{\frac{p}{q} = B\rho} \quad \text{or} \quad \boxed{\frac{p [\text{GeV}/c]}{\rho [\text{m}]} = 0.2998 B [\text{T}]}$$

- ▶ The ratio of p to q describes the “stiffness” of a beam, it can be considered as a measure of how much angular deflection results when a particle travels through a given magnetic field
 - ▶ For a specific magnetic field, the greater the momentum of a particle, the less it will be bent as it travels through that field
 - ▶ The greater the charge of a particle, the more it will be bent as it travels through a given magnetic field

Transfer matrix of a sector dipole

- ▶ For a dipole magnet the linear transfer matrix for (x, x') looks like this:

$$\begin{pmatrix} x \\ x' \end{pmatrix}_f = \begin{pmatrix} \cos \phi & \rho \sin \phi \\ -\frac{1}{\rho} \sin \phi & \cos \phi \end{pmatrix} \begin{pmatrix} x \\ x' \end{pmatrix}_i$$

Here $\phi = l/\rho$ (length and bending radius)

- ▶ The matrix taking into account $\Delta p/p_0$ is a bit more complicated than that:

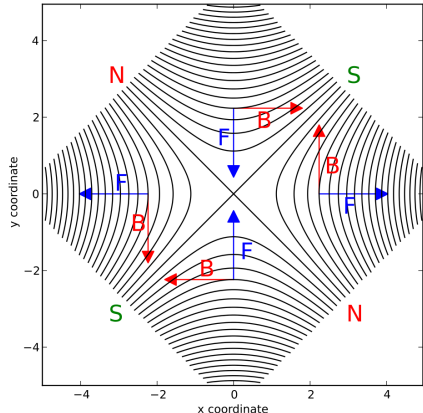
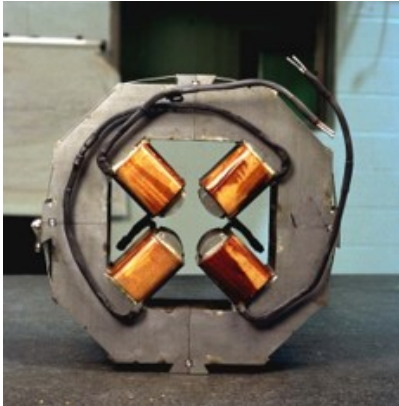
$$\begin{pmatrix} x \\ x' \\ \Delta p/p_0 \end{pmatrix}_f = \begin{pmatrix} \cos \phi & \rho \sin \phi & \rho(1 - \cos \phi) \\ -\frac{1}{\rho} \sin \phi & \cos \phi & \sin \phi \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ x' \\ \Delta p/p_0 \end{pmatrix}_i$$

- ▶ The vertical component is a drift:

$$\begin{pmatrix} y \\ y' \end{pmatrix}_f = \begin{pmatrix} 1 & l \\ 0 & 1 \end{pmatrix} \begin{pmatrix} y \\ y' \end{pmatrix}_i$$

Focusing particles: quadrupole magnet

- ▶ Quadrupole field provides focusing/defocusing force in the transverse direction
 - ▶ ...much needed improvement to the dipole-only field we discussed above
- ▶ There is a catch though: it focuses in one direction, defocuses in the other



Transfer matrix for a *horizontally focusing* quadrupole magnet

- ▶ For a *horizontally focusing* quadrupole magnet the linear transfer matrix for (x, x') looks like this:

$$\begin{pmatrix} x \\ x' \end{pmatrix}_f = \begin{pmatrix} \cos \phi & \frac{1}{\sqrt{|k|}} \sin \phi \\ -\sqrt{|k|} \sin \phi & \cos \phi \end{pmatrix} \begin{pmatrix} x \\ x' \end{pmatrix}_i$$

Here $\phi = l\sqrt{|k|}$ (length and magnet strength), $k < 0$ for the horizontally focusing magnet

- ▶ The vertical component is defocusing:

$$\begin{pmatrix} y \\ y' \end{pmatrix}_f = \begin{pmatrix} \cosh \phi & \frac{1}{\sqrt{|k|}} \sinh \phi \\ \sqrt{|k|} \sinh \phi & \cosh \phi \end{pmatrix} \begin{pmatrix} y \\ y' \end{pmatrix}_i$$

- ▶ For $k > 0$ the magnet is vertically focusing and horizontally defocusing, so the M_x and M_y matrices are interchanged

Thin-lens approximation

- ▶ Assuming $f = 1/(kl) \gg l$, the transfer matrices can be approximated as

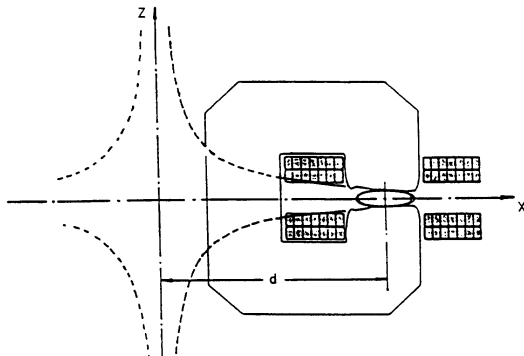
$$\begin{pmatrix} x \\ x' \end{pmatrix}_f = \begin{pmatrix} 1 & 0 \\ -\frac{1}{f} & 1 \end{pmatrix} \begin{pmatrix} x \\ x' \end{pmatrix}_i$$

and

$$\begin{pmatrix} y \\ y' \end{pmatrix}_f = \begin{pmatrix} 1 & 0 \\ \frac{1}{f} & 1 \end{pmatrix} \begin{pmatrix} y \\ y' \end{pmatrix}_i$$

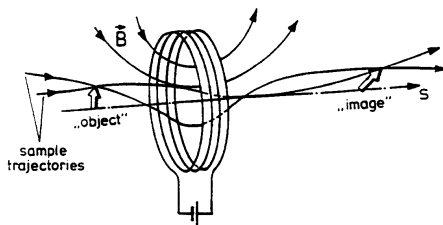
- ▶ Look familiar?

Combined-function magnet (Synchrotron magnet)



- ▶ Combines the homogeneous field of a dipole with a quadrupole gradient
- ▶ Potential: $V(x, y) = B_0 y - gxy$
- ▶ Advantage: compact lattice
- ▶ Disadvantage: strong correlation between momentum and beam optics, poor flexibility
- ▶ Synchrotron magnet behaves as a shifted quadrupole (see figure)

Solenoid

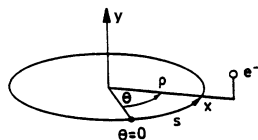


- ▶ Simultaneous focusing/defocusing in all directions (compare to quadrupoles)
- ▶ Helical motion $\Rightarrow$ coupling between the two transverse planes (use carefully)
- ▶ Long solenoids work well for particle transport (mind the fringe fields, $\nabla \cdot \mathbf{B} = 0$ implies that magnetic field lines close on themselves)
- ▶ Short solenoids could be used for focusing but only at low momenta:

$$\frac{1}{f_{sol}} = \int \left(\frac{qB_{sol}}{2p} \right)^2 ds$$

hence, f_{sol} scales as p^2

Weak focusing I



- ▶ Consider a magnetic field independent of the azimuthal angle θ
- ▶ On the reference orbit
$$F_{\text{Lorentz}} = F_{\text{centrifugal}} (p/q = B\rho)$$
- ▶ Stability: small deviations from the ideal orbit $\Rightarrow$ restoring forces

- ▶ That implies:

$$qvB_y(r) = \begin{cases} < \frac{mv^2}{r} & \text{for } r < \rho \\ > \frac{mv^2}{r} & \text{for } r > \rho \end{cases}$$

$$\text{or } F_{\text{rest}} = \frac{mv^2}{r} - qvB_y$$

- ▶ Constant B field provides stability only in the horizontal plane (any vertical displacement results in a particle spiraling away from the reference orbit plane)
- ▶ Need field gradient

Weak focusing II



$$F_{\text{rest}} = \frac{mv^2}{r} - qvB_y$$

- ▶ Introduce field gradient:

$$qvB_y = qv \left(B_0 + \frac{\partial B_y}{\partial r} x \right) = qvB_0 \left(1 + \frac{\partial B_y}{\partial r} \frac{x}{B_0} \right) = qvB_0 \left(1 - n \frac{x}{\rho} \right),$$

where $n \stackrel{\text{def}}{=} -\frac{\rho}{B_0} \frac{\partial B_y}{\partial r}$ is the *field index*, ρ the reference orbit radius

- ▶ For small deviation from the reference orbit x we have
 $r = \rho + x = \rho \left(1 + \frac{x}{\rho} \right)$ and $\frac{mv^2}{r} = \frac{mv^2}{\rho(1+x/\rho)} \approx \frac{mv^2}{\rho} \left(1 - \frac{x}{\rho} \right)$

- ▶ Thus, we can rewrite the expression for F_{rest} :

$$F_{\text{rest}} = \frac{mv^2}{\rho} \left(1 - \frac{x}{\rho} \right) - qvB_0 \left(1 - \frac{nx}{\rho} \right) = -qvB_0 \frac{x}{\rho} (1 - n)$$

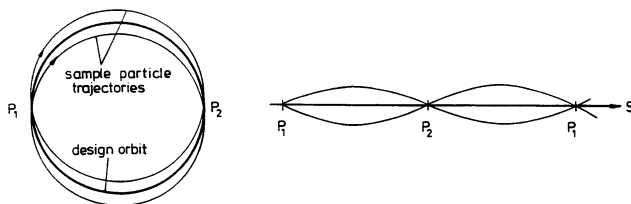
- ▶ For focusing in the horizontal plane we need $n < 1$

Weak focusing III

- From the previous page:

$$F_{\text{rest}} = -qvB_0 \frac{x}{\rho}(1 - n), \quad n < 1$$

- This does not exclude $n = 0$, so, as mentioned above, a pure dipole provides stability in the horizontal plane. However,



- Huge amplitude of betatron oscillations: maximum orbit deviation is $\alpha\rho$, where α is the initial beam divergence.
 - $\alpha = 1 \text{ mrad}$, $\rho = 1000 \text{ m} \Rightarrow$ orbit deviation of 1 m

Weak focusing in the vertical plane (last)

- ▶ We need a restoring force in the y direction: $F_y \propto -y$
- ▶ Need a horizontal magnetic field component: $B_x = -Cy$
- ▶ Or a negative horizontal field gradient: $\frac{\partial B_x}{\partial y} = -C$
- ▶ Maxwell's equation: $\nabla \times \mathbf{B} = 0$ (no sources within the aperture), so

$$\frac{\partial B_x}{\partial y} = \frac{\partial B_y}{\partial x} = \frac{\partial B_y}{\partial r} < 0$$

- ▶ Thus, the vertical field component has to decrease with increasing radius: $\frac{\partial B_y}{\partial r} < 0 \Leftrightarrow n > 0$
- ▶ Resulting equations of motion (without derivation, see notes):
 $\ddot{x} + \omega_0^2(1 - n)x = 0$ and $\ddot{y} + \omega_0^2 ny = 0$
- ▶ With $0 < n < 1$ there is always less than one oscillation per turn $\Rightarrow$ large oscillation amplitudes
- ▶ Strong focusing to the rescue!

Strong focusing

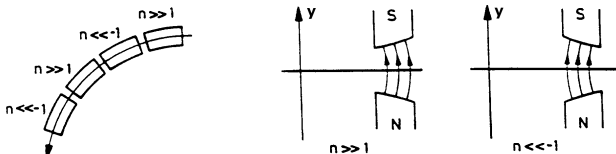


Figure 5: Alternating-gradient focusing

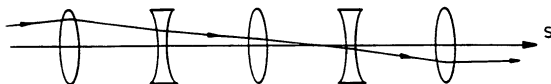


Figure 6: An alternating series of focusing and defocusing lenses leads to overall focusing if the distances between the lenses are not too large.

- Break azimuthal symmetry
- Use a sequence of focusing and defocusing magnets with $n \gg 1$
- If we are careful, the net effect is focusing!